Assessing Cybersecurity Awareness and Online Safety Practices Among High School and University Students in Burkina Faso

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Abstract

The rapid spread of internet access, smartphones, and digital technologies has reshaped education and communication among young people in Burkina Faso. This growing connectivity, however, has also exposed students to a wider range of cybersecurity threats, including phishing, malware, identity theft, online scams, and social engineering. Despite these risks, very little empirical research has looked specifically at how aware Burkinabè students actually are of these threats.

This study assesses cybersecurity awareness and online safety practices among high school and university students in Burkina Faso through an exploratory quantitative design. Data were collected using an online questionnaire built on Google Forms, completed by 33 students. Descriptive statistics were used to analyze participants' knowledge of cybersecurity concepts, password habits, phishing awareness, online privacy behavior, and general attitudes toward cybersecurity.

The results show that while most participants had at least heard of cybersecurity, real gaps remain in both knowledge and behavior: a large share of students reported unsafe practices such as password reuse, occasional interaction with suspicious links, and inconsistent verification of website authenticity. At the same time, almost all participants expressed a strong willingness to learn more about the subject. The study concludes that cybersecurity education needs to be strengthened in schools and universities across Burkina Faso, and offers practical recommendations for educators, policymakers, and the development of digital learning platforms such as CYBERIA.

Keywords: cybersecurity awareness, online safety, students, digital literacy, Burkina Faso, cybersecurity education.

Contents

1 Introduction

1.1 Background of the Study

The rapid growth of internet access, smartphones, and digital technologies has changed the way people learn, communicate, and access information all over the world. Burkina Faso is no exception: students increasingly depend on digital devices and online platforms for schoolwork, communication, entertainment, and social networking. These tools open up real educational and social opportunities, but they also expose young users to a wide range of cybersecurity threats.

Phishing attacks, malware infections, identity theft, online scams, cyberbullying, and social engineering are becoming more common, and students are particularly vulnerable. Many of them use online services daily without having the cybersecurity knowledge or habits needed to stay safe.

Cybersecurity awareness can be described as a person's ability to recognize cyber threats, understand what they could lead to, and adopt behaviors that protect their devices, personal information, and online accounts. Research consistently shows that raising this kind of awareness reduces the likelihood of cyber incidents and encourages safer digital behavior. Some African countries have started integrating cybersecurity awareness into their education systems, but in Burkina Faso this remains limited in most schools and universities.

1.2 Problem Statement

Students in Burkina Faso are relying more and more on digital technologies for learning and communication, and this dependence comes with growing exposure to cyber threats. Yet very little empirical evidence exists on how aware high school and university students actually are of these risks. Without this kind of information, schools and policymakers have no solid basis for designing awareness campaigns or building cybersecurity content into the curriculum. This study addresses that gap by examining cybersecurity awareness and online safety practices among students in Burkina Faso.

1.3 Research Questions

Main research question: What is the level of cybersecurity awareness and online safety practices among high school and university students in Burkina Faso?

Specific research questions:

1. What do students know about common cybersecurity threats?
2. What online safety practices do students actually adopt?
3. How do students perceive cybersecurity risks while using the Internet?
4. What factors influence students' cybersecurity awareness?
5. What can educational institutions do to strengthen cybersecurity awareness?

1.4 Research Objectives

The general objective is to assess cybersecurity awareness and online safety practices among high school and university students in Burkina Faso. The specific objectives are to: evaluate students' knowledge of common cybersecurity threats; examine their online safety practices; identify risky online behaviors;

determine the factors that influence cybersecurity awareness; and propose practical recommendations for improving cybersecurity education in Burkina Faso.

1.5 Scope and Significance of the Study

This research focuses on cybersecurity awareness and online safety practices among high school and university students in Burkina Faso, covering password security, phishing awareness, malware, social engineering, online privacy, and general safe internet practices. It is limited to students who voluntarily completed the online questionnaire; teachers, government employees, and cybersecurity professionals were not included.

The study adds to the growing body of knowledge on cybersecurity awareness in Africa by offering one of the first exploratory studies focused specifically on students in Burkina Faso. Its findings can help schools and universities design more targeted awareness programs, support policymakers working to integrate cybersecurity education into school curricula, and provide direct evidence for the continued development of CYBERIA, an educational platform built to improve cybersecurity knowledge through interactive lessons, quizzes, and online safety training.

2 Literature Review

2.1 Concept of Cybersecurity Awareness

Cybersecurity awareness refers to the knowledge, attitudes, and behaviors that allow individuals to recognize cyber threats and adopt safe online practices. It involves understanding common cyber risks, spotting malicious activity, and applying the right security measures to protect personal information and digital assets. According to the National Institute of Standards and Technology (NIST, 2024), cybersecurity awareness is a critical part of cybersecurity as a whole, because human behavior remains one of the leading causes of security incidents. Students are among the most active users of digital technology, which makes cybersecurity awareness essential for protecting both their personal and academic information.

2.2 Common Cybersecurity Threats Affecting Students

Students encounter a range of cybersecurity threats in their daily use of the internet.

Phishing is a fraudulent attempt to obtain sensitive information – usernames, passwords, or financial details – by posing as a trusted individual or organization, typically through emails, social media messages, and fake websites.

Malware is malicious software designed to damage, steal from, or disrupt digital systems, including viruses, worms, ransomware, spyware, and trojans.

Social engineering involves manipulating people into revealing confidential information or taking actions that compromise their own security, exploiting human psychology and trust rather than a technical flaw.

Weak password practices remain common: many students use simple, predictable passwords and reuse the same password across several accounts, considerably increasing their exposure to attacks.

2.3 Cybersecurity Awareness Among Students in Africa

Cybersecurity awareness has become an increasingly active research topic across Africa as internet penetration and smartphone adoption continue to rise. Eltahir and Ahmed (2023), studying undergraduate students in Sudan, found generally low cybersecurity awareness despite frequent internet use, and argued that universities should build structured cybersecurity education into their curricula. Research in South Africa found a clear gap between how capable students believed themselves to be online and how they actually behaved: many students who felt confident about protecting themselves still reused passwords, clicked on suspicious links, and managed their personal information carelessly. Studies conducted in Nigeria reported moderate awareness among university students, most of whom recognized basic cybersecurity concepts but lacked the practical skills needed to identify phishing attempts and other forms of cybercrime. Taken together, these findings suggest that cybersecurity awareness among African students remains uneven.

2.4 Cybersecurity Education in Educational Institutions

Schools and universities play a central role in preparing students to use digital technology safely and responsibly, yet cybersecurity education remains limited across much of Africa. In most cases, cybersecurity topics appear only within computer science or IT programs, leaving students in other fields with little formal exposure. Some universities have introduced awareness campaigns, seminars, and workshops, often timed around events such as Cybersecurity Awareness Month, but these tend to be one-off events rather than sustained programs. Researchers generally recommend building cybersecurity into national curricula starting from secondary school.

2.5 Factors Influencing Cybersecurity Awareness

Previous studies point to several factors that shape students' cybersecurity awareness, including age, gender, educational level, frequency of internet use, access to smartphones and computers, previous cybersecurity training, digital literacy level, and field of study. Students who have received formal cybersecurity training generally show safer online behavior than those who have not; however, awareness alone does not automatically translate into secure behavior.

2.6 Research Gap

Cybersecurity awareness has already been studied in countries such as Sudan, South Africa, Nigeria, and Kenya, but very little research has focused specifically on Burkina Faso. Most existing studies also concentrate on university students and rarely include high school students, and few look at knowledge, attitudes, and actual online behavior together in the same study. The present study addresses this gap directly by offering an exploratory assessment of cybersecurity awareness among both high school and university students in Burkina Faso.

3 Theoretical and Conceptual Framework

3.1 Theoretical Framework: Protection Motivation Theory

This study draws on Protection Motivation Theory (PMT), developed by Rogers (1975), which explains how individuals adopt protective behaviors when faced with a perceived threat. According to PMT, people weigh both the severity of a threat and their own ability to respond to it before deciding whether to act. The theory identifies four components: *perceived severity* (the belief that cyber threats can cause real harm), *perceived vulnerability* (the belief that one is personally at risk), *response efficacy* (the belief that cybersecurity measures actually reduce risk), and *self-efficacy* (confidence in one's own ability to apply safe practices).

Applied to this research, PMT offers a useful lens for understanding how students' perceptions of cyber threats shape their online behavior. Students who see cyber threats as serious and believe they can protect themselves are more likely to use strong passwords, avoid suspicious links, and generally guard their personal information.

3.2 Conceptual Framework

Building on this theoretical grounding, the conceptual framework sets out the relationship between students' demographic characteristics, their cybersecurity awareness, and their online safety practices, as summarized in Table 1.

Table 1: Conceptual framework of the study

Independent variables	**Mediating variable**	**Dependent variable**
Age, gender, educational level, internet usage frequency, device ownership, previous cybersecurity training, digital literacy	Cybersecurity awareness	Online safety practices: password management, phishing detection, website verification, privacy protection, safe social media use

The framework assumes that students' demographic characteristics and prior experiences shape their level of cybersecurity awareness, which in turn affects how safely they behave online.

4 Research Methodology

4.1 Research Design

The study used a quantitative descriptive research design to assess cybersecurity awareness and online safety practices among high school and university students in Burkina Faso. A quantitative approach was chosen because it makes it possible to collect numerical data from a group of participants and identify patterns using descriptive statistics.

4.2 Study Area and Target Population

The study was carried out in Burkina Faso, a West African country experiencing rapid growth in internet access, smartphone usage, and digital services. The target population consisted of high school and university students currently enrolled in educational institutions in Burkina Faso with access to the internet.

4.3 Sampling Technique and Sample Size

A convenience sampling technique was used: participants completed the online questionnaire voluntarily after receiving the survey link through WhatsApp, Facebook, LinkedIn, and other digital platforms. A total of 33 students took part, including both high school and university students. The sample is relatively small, but it offers useful preliminary insight into cybersecurity awareness among students in the country.

4.4 Data Collection Instrument

Data were collected through a structured online questionnaire built with Google Forms, organized into five sections: (A) demographic information (age, gender, educational level); (B) internet usage (frequency, devices, social media use); (C) cybersecurity knowledge (understanding of cybersecurity, phishing, malware, password security, online privacy); (D) online safety practices (password management, website verification, link-clicking behavior, information sharing, privacy settings); and (E) attitudes toward cybersecurity.

4.5 Data Collection Procedure

The questionnaire was distributed electronically through WhatsApp groups, Facebook, LinkedIn, and personal contacts. Participation was entirely voluntary, and participants were informed of the study's purpose and assured of anonymity beforehand. Data collection lasted about one week, yielding 33 valid responses.

4.6 Data Analysis

The collected data were exported from Google Forms into Microsoft Excel for analysis. Descriptive statistical techniques – frequencies, percentages, tables, and charts – were used to summarize participants' cybersecurity awareness and online safety practices. No inferential statistical tests were conducted, given the exploratory nature of the study and the relatively small sample size.

4.7 Reliability, Validity, and Ethical Considerations

All participants answered the same standardized questionnaire under identical conditions, and the instrument was developed after reviewing previous cybersecurity awareness studies. Content validity was supported by aligning the questionnaire with the study's objectives and key cybersecurity topics. Ethically, participation was voluntary, respondents could withdraw at any time, no personally identifiable information was collected, and all responses were treated confidentially and analyzed in aggregate form.

5 Results and Discussion

5.1 Demographic Characteristics of Participants

A total of 33 students took part in the study. Most (90.9%) were between 15 and 18 years old, 6.1% were between 19 and 22, and 3.0% were 23 or older, meaning the study mainly reflects the views of high school students. Gender representation was 57.6% male, 39.4% female, and 3.0% preferred not to say. Educational level was 87.9% high school, 6.1% university, and 6.0% other, as summarized in Table 2.

5.2 Internet Usage

Internet use among respondents was consistently high: 81.8% reported going online several times a day. The smartphone was by far the dominant device (87.9%), followed by tablets (9.1%) and laptops (3.0%),

Table 2: Demographic characteristics of respondents ($n = 33$)

Characteristic	**Category**	**n**	**%**
Age	15–18 years	30	90.9%
	19–22 years	2	6.1%
	23 years or older	1	3.0%
Gender	Male	19	57.6%
	Female	13	39.4%
	Prefer not to say	1	3.0%
Educational level	High school	29	87.9%
	University	2	6.1%
	Other	2	6.0%

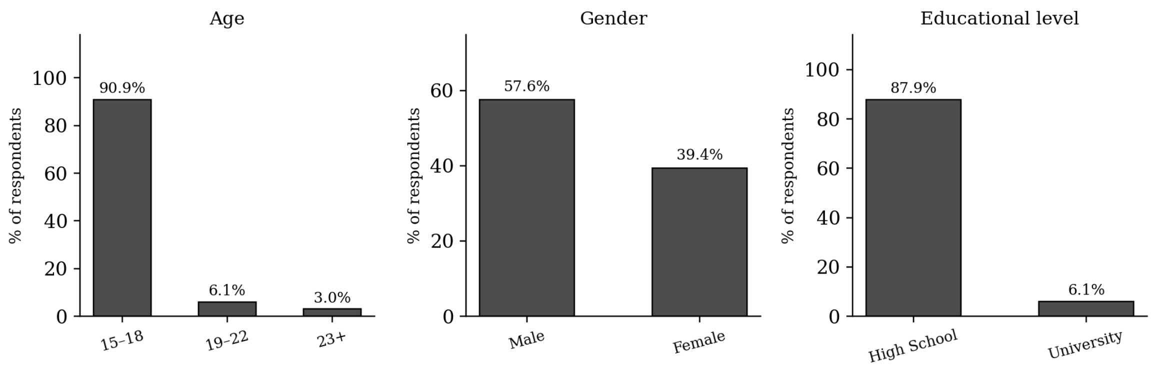

Figure 1: Distribution of respondents by age, gender, and educational level.

suggesting that cybersecurity awareness efforts should be designed primarily around smartphone use.

5.3 Cybersecurity Knowledge

Most respondents (93.9%) said they had heard of cybersecurity before, but their grasp of specific concepts was less consistent. Only 60.6% correctly identified phishing as an online scam designed to steal sensitive information, and 66.7% correctly recognized malware as malicious software (Table 3). Students are familiar with the word “cybersecurity,” but their understanding of specific, common threats is only moderate.

Table 3: Cybersecurity knowledge among respondents

Concept	**Correctly identified**
Has heard of cybersecurity	93.9%
Correctly defines phishing	60.6%
Correctly defines malware	66.7%

5.4 Online Safety Practices

Reported behavior was mixed. Only 45.5% of respondents said they always create strong passwords, and 39.4% acknowledged reusing the same password across multiple accounts. Only 30.3% said they never click on unknown links, and only 21.2% said they always verify a website’s authenticity before entering personal information (Table 4). These figures point to risky habits that leave students more exposed to cyber threats than their stated awareness of cybersecurity would suggest.

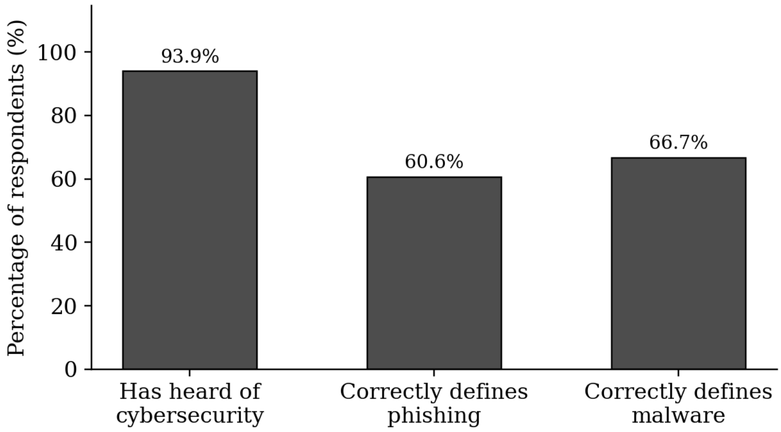

Figure 2: Cybersecurity knowledge among respondents.

Table 4: Online safety practices

Practice	Share reporting safe behavior
Creates strong passwords (always)	45.5%
Reuses passwords across accounts (admits reuse)	39.4%
Never clicks unknown links	30.3%
Always verifies website authenticity	21.2%

5.5 Experience with Cyber Incidents and Social Media Practices

Among all participants, 30.3% said they had previously experienced an online scam or hacking incident, while 69.7% had not. On social media, many respondents said they share personal information online either frequently or occasionally, and a large share admitted to accepting friend requests from strangers. Regarding privacy settings specifically, 54.5% said they adjust them regularly, while 45.5% rarely or never do.

5.6 Self-Perceived Safety and Interest in Cybersecurity Education

Respondents were notably cautious when asked to rate their own online safety directly. Only 36.4% agreed with the statement “I believe I am safe online,” 45.5% were neutral, and 18.2% disagreed or strongly disagreed. In other words, most students do not confidently see themselves as safe online – a level of uncertainty that fits the knowledge and behavioral gaps identified above, and that is consistent with Protection Motivation Theory: without strong self-efficacy or confidence in protective measures, students are less likely to adopt consistent safe practices.

At the same time, interest in learning more was almost universal: 32 of the 33 respondents (97.0%) said they would like to learn more about cybersecurity, and only one participant said they were not interested (Figure 4). This combination – moderate self-confidence but very high willingness to learn – represents a clear opening for schools and universities to introduce structured, practical cybersecurity awareness programs.

5.7 Discussion

These findings line up closely with previous research from other African countries. Eltahir and Ahmed (2023) found limited cybersecurity awareness among Sudanese university students despite frequent in-

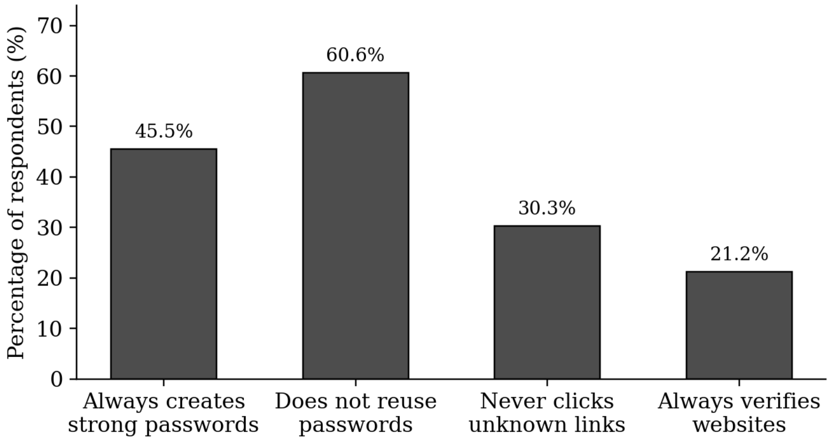

Figure 3: Share of respondents reporting the safer behavior for each online safety practice.

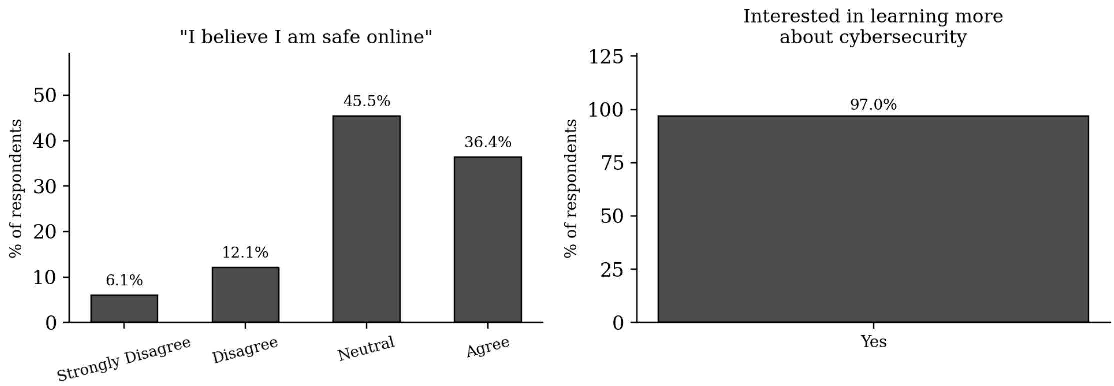

Figure 4: Self-perceived online safety and interest in learning more about cybersecurity.

ternet use; studies in South Africa found that students often overestimated their own cybersecurity abilities while continuing risky practices; and research in Nigeria likewise found moderate awareness alongside clear gaps in phishing detection and password management. The present study reveals a comparable pattern among students in Burkina Faso: nearly all respondents had heard of cybersecurity, yet many still reused passwords, verified websites inconsistently, and interacted with suspicious links. Interestingly, unlike the South African students who tended to overestimate their own cybersecurity abilities, most respondents in this study were neutral or uncertain about their own online safety – suggesting that Burkinabè students may be more aware of their own limitations, even if this awareness does not yet translate into consistently safe behavior. What stands out is the exceptionally high interest in learning more – an opportunity that schools and universities in Burkina Faso are well placed to act on. Overall, improving cybersecurity awareness among students should not stop at raising knowledge levels; it needs to actively encourage long-term behavioral change through practical training and sustained awareness campaigns.

6 Recommendations

6.1 For Educational Institutions

Schools and universities should integrate cybersecurity education into their curricula starting from secondary school, organize regular awareness campaigns and hands-on training covering phishing detection, password management, online privacy, and social engineering, and ensure teachers themselves receive cybersecurity training.

6.2 For Students

Students can meaningfully reduce their exposure to cyber threats by creating strong, unique passwords for every account; enabling multi-factor authentication wherever available; being cautious with unfamiliar emails, messages, and links; keeping devices and applications up to date; verifying a website's authenticity before submitting personal information; being deliberate about what they share on social media; and taking part in cybersecurity awareness programs when available.

6.3 For Policymakers

Government agencies and education authorities should strengthen national digital literacy initiatives by building cybersecurity education into secondary and higher education curricula, work with universities, schools, and technology partners to run nationwide awareness campaigns, and prioritize investment in teacher training and digital infrastructure.

6.4 For Future Researchers

Future studies should aim for a larger, more representative sample covering different regions of Burkina Faso and extend the research to other groups, including teachers, parents, government employees, small business owners, rural communities, and primary school students. It would also be valuable to evaluate the effectiveness of cybersecurity awareness programs directly, by comparing students' knowledge before and after training.

7 Practical Implications: The CYBERIA Platform

Beyond its academic contribution, this study has direct practical value. Educational institutions can use its results to identify specific weaknesses in students' cybersecurity knowledge and design targeted awareness programs around them. The findings also directly inform the continued development of CYBERIA, an educational platform dedicated to digital literacy and cybersecurity education. Planned features for future versions of CYBERIA include interactive cybersecurity lessons, phishing simulations, password security training, online safety quizzes, digital certificates, practical cybersecurity challenges, and AI-powered learning assistants – turning the research findings into concrete tools that improve students' online safety in practice.

8 Limitations and Future Work

A few limitations should be kept in mind when interpreting these findings. The study involved only 33 participants, which limits how far the results can be generalized to the broader student population of

Burkina Faso, and most respondents were high school students, so university students are only lightly represented. The data were self-reported, which leaves room for recall bias or socially desirable answers, and convenience sampling may have introduced some bias.

Future work should expand this research to larger and more diverse populations across Burkina Faso, and examine the effectiveness of cybersecurity awareness interventions through experimental or longitudinal designs. Related topics worth exploring include artificial intelligence and cybersecurity awareness, mobile security practices, cyberbullying among students, digital citizenship, online fraud awareness, and privacy protection on social media. This research will also continue to inform the development of CYBERIA as a digital learning platform, with the goal of making cybersecurity education accessible to students across Burkina Faso and, eventually, more broadly across Africa.

9 Conclusion

This study set out to assess cybersecurity awareness and online safety practices among high school and university students in Burkina Faso. The findings show that students are highly connected to the internet and generally familiar with the idea of cybersecurity, but real knowledge gaps and unsafe habits persist – password reuse, inconsistent website verification, and uneven privacy management continue to expose them to unnecessary risk. At the same time, the study found a genuinely encouraging result: almost all participants said they wanted to learn more about cybersecurity, showing that students already recognize the importance of online safety and are ready to build their knowledge further.

Despite its modest sample size, this study offers valuable preliminary evidence on cybersecurity awareness among Burkinabè students and adds to the still-limited body of cybersecurity research conducted within the country. Its findings have already informed the development and planned expansion of CYBERIA, an educational platform designed to build cybersecurity awareness through interactive lessons, quizzes, simulations, and other digital learning resources. By combining academic research with practical educational innovation, Burkina Faso can help build a safer, more digitally resilient generation of students.

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A Survey Questionnaire

The complete Google Forms questionnaire used for data collection is available upon request and will be inserted here in the final submitted version.

B Google Forms Summary Charts

The key summary charts (age, gender, educational level, cybersecurity knowledge, online safety practices, self-perceived safety, and interest in learning more) are presented directly in Section 5 alongside the corresponding results. The raw Google Forms export (individual response data) is available upon request.

C The CYBERIA Educational Platform

As an extension of this research, the CYBERIA educational platform is being developed to improve cybersecurity awareness among students through interactive learning resources, quizzes, practical exercises, phishing simulations, and digital certifications. The platform aims to translate the findings of this study into practical educational tools that promote safer online behavior among students in Burkina Faso and, over time, across Africa.